

Code:

```
#include <stdio.h>

int gcd(int a, int b) {
    while (b) {
        int t = a % b;
        a = b;
        b = t;
    }
    return a;
}

int sumHexDigits(int x) {
    int sum = 0;
    while (x > 0) {
        sum += x % 16;
        x /= 16;
    }
    return sum;
}

int search(int a, int b) {
    int count = 0;
    for (int x = a; x <= b; x++) {
        if (gcd(x, sumHexDigits(x)) > 1) {
            count++;
        }
    }
    return count;
}

int main() {
    int T;
    if (scanf("%d", &T) == 1) {
        while (T--) {
            int L, R;
            scanf("%d %d", &L, &R);
            printf("%d\n", search(L, R));
        }
    }
    return 0;
}
```

```
}
```

Input:

```
3
1 3
5 8
7 12
```

Output:

```
2
4
6
```